MIL-C-48397A (AR)
22 OCTOBER 1986
SUPERSEDING
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17 March 1981

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M752A1
LOADING, ASSEMBLING AND PACKING

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is avaiable for use by all departments and agencies for the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements and provides quality assurance provisions for the loading, assembling and packing of the M752A1, Ignition Cartridge for 81MM M853 Illuminating and M819 Smoke Cartridges.

2. APPLICABLE DOCUMENTS

2.1 Government documents

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-223	- Powder, Black
MIL-P-46425	- Primer, Percussion, M35, Loading Assembling and Packing
MIL-A-48078	- Ammunition, Standard Quality Assurance Provisions, General Specification for
MIL-P-60398	- Propellant M9, Flake (For use in Ignition Cartridges)

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, U.S. Army Armament Research, Development and Engineering Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286	- Propellants, Solid, Sampling, Examination and Testing
MIL-STD-414	- Sampling Procedures and Tables for Inspection by Variables for Percent Defective

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS (See 6.9)

US ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

PRODUCT AND PACKAGING DRAWINGS

9278622	- Envelope Drawing for Leak Testing of Ignition Cartridges
9294752	- Cartridge, Ignition, M752A1, Loading Assembly
9294949	- Packing, and Marking for Box Wirebound, Packing for Cartridge Ignition - M752A1
9294950	- Box, Fiberboard
9294951	- Box, Wirebound
9294952	- Tray assembly for Cartridge, Ignition M752A1

INSPECTION EQUIPMENT DRAWINGS

9278622	- Envelope Drawing for Leak Testing of Ignition Cartridges

CODE OF FEDERAL REGULATIONS

49 CFR 71-90	- Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

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(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with applicable drawings and specifications.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing (dwg.) 9294752 and with all requirements specified in applicable specifications, standards, and associated drawings.

3.3 Moisture content and black powder pellet. The moisture content of the black powder pellet (dwg. 7549177) at the time of loading shall not exceed 0.50 percent. Testing shall be as specified in 4.5.2.

3.4 Percussion primer sensitivity

3.4.1 Non-functioning. The primer shall not function under an applied energy value of 8.85 inch ounces minimum (min.) when tested as specified in 4.5.3.1.

3.4.2 Functioning. The primer shall function under an applied energy value of 82.60 inch ounces maximum (max.) when tested as specified 4.5.3.2.

3.4.3 Safety drop test. The primer shall not function as a result of the five foot cartridge drop specified in 4.5.3.3.

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3.5 Leak test. The loaded cartridges shall satisfactorily pass the leak test specified in 4.5.4.

3.6 Proving ground test. The ignition cartridges shall comply with the following requirements (see 4.5.6).

3.6.1 Misfires. The ignition cartridges shall function on the first attempt, without any audible hangfire.

3.6.2 Velocity and functioning. The cartridge assemblies shall function and shall impart a corrected muzzle velocity of not more than 215 feet per second (fps) and not less than 195 fps. In addition, the velocity standard deviation of the lot shall be no greater than 3.1 fps.

3.6.3 Metal parts security. There shall be no evidence of rupture or metal parts separation in the gun bore or in flight.

3.7 Propellant charge uniformity. The M9 propellant lots shall achieve a velocity of 205 fps (see 4.5.7). This test shall be required when a lapse of six months or more occurs between propellant acceptance and LAP initiation.

3.8 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions (D.I.D.'s) cited in 4.4.3.2.1, 4.5.2 and 4.5.4.

3.9 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specifed in the contract.

3.10 Workmanship. All parts and assemblies shall be fabricated and loaded in a thorough workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, rust and other foreign matter. The cleaning methods used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the

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performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made part of this detail specification.

4.1.1 Responsibility for compliance. All items must meet all requirements of section 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The following types of inspection shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

4.3 First Article Inspection

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the items in the quantities indicated in Table I.

4.3.2 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.3 Rejection. See MIL-A-48078.

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TABLE I. First article inspection

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Flash Tube Assembly (dwg. 9294757)				
	Examination for Defects	25	100%	3.2	4.4.2.1
	Tube Assembly (dwg. 9294753)				
	Examination for Defects	25	100%	3.2	4.4.2.2
	Body Assembly (dwg. 12630551)				
	Examination for Defects	25	100%	3.2	4.4.2.3
	Push-out test	10	100%	3.2	4.4.3.1/4.5.1
	Cartridge, Ignition Prior to Filling with Propellant (dwg. 9294752)				
	Examination for Defects	25	100%	3.2	4.4.2.5
	Sensitivity Non-functioning	25	100%	3.4.1	4.5.3.1
	Sensitivity Functioning	25	100%	3.4.2	4.5.3.2
	Safety Drop Test	25	100%	3.4.3	4.5.3.3
	Cartridge, Ignition, After Assembly of Cap (dwg. 9294752)				
	Examination for Defects	25	100%	3.2	4.4.2.7
	Leak Test	50	100%	3.5	4.5.4
	Proving Ground Test	50	100%	3.6	4.5.6

NOTES:

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4.4 Quality conformance inspection

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of ignition cartridges shall contain:

a. Metal parts of one lot interfix from one manufacturer.

b. Black powder from not more than one lot from one manufacturer.

c. Percussion elements from not more than one lot from one manufacturer.

d. M9 propellant from not more than one lot from one manufacturer.

e. Heat shrinkable tubing from not more than one manufacturer or manufacturer part number.

f. Adhesive sealant from one manufacturer.

g. Tubes from one lot interfix.

4.4.2 Examination. See MIL-A-48078

a. Sampling plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9294751
4.4.2.1	Flash Tube Assembly				NEXT HIGHER ASSEMBLY 9294752
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	REFERENCE PARAGRAPH / INSPECTION METHOD
Critical					
1	Heat shrinkable tubing missing, loose or does not completely cover all flash holes		100%	3.2	Visual/Manual
2	Pin missing		100%	3.2	Visual
Major					
101	Length, max.		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual
NOTES:					

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QUALITY COMFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.2	Tube Assembly				9294753
					NEXT HIGHER ASSEMBLY 9294752
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter, max.		0.40%	3.2	Gage
102	Inside diameter, min.		0.40%	3.2	Gage
103	Tubing missing, loose, wrinkled, separating or inadequately covering external surface		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.3	Body Assembly				12630551
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Primer missing or not firmly seated		0.40%	3.2	Visual/Manual
102	Push out test	a	100%	3.2	4.4.3.1/4.5.1
103	Lacquer seal missing or inadequate		0.40%	3.2	Visual
104	Presence of lacquer on face of primer		0.40%	3.2	Visual
105	Distance, max, from face of primer to shoulder		0.40%	3.2	Gage
106	Flatness of primer face after swaging		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: (a) Five samples randomly selected from each eight hours production.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48397A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9549177
4.4.2.4	Pellet				NEXT HIGHER ASSEMBLY 9294752
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight of pellet		0.40%	3.2	Balance
102	Moisture content	25(a)	100%	3.3	4.5.2
103	Outside diameter		0.40%	3.2	Gage
104	Inside diameter		0.40%	3.2	Gage
105	Length		0.40%	3.2	Gage
106	Pellet broken		0.40%	3.2	Gage
Minor					
201	Pellet chipped or cracked		0.65%	3.1	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: (a) Twenty-five randomly selected pellets from each eight hours production.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48397A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.5	Cartridge, Ignition Prior to Filling with Propellant				9294752
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical					
1	Firing plug not loose after assembly		100%	3.2	Visual/Manual
2	Flash tube assembly missing		100%	3.2	Visual
3	Sensitivity non-functioning		(a)	3.4.1	4.5.3.1
4	Safety drop test	25	100%	3.4.3	4.5.3.3
Major					
101	Percussion primer inverted or missing		100%	3.2	Visual
102	Lacquer seal missing or inadequate		0.40%	3.2	Visual
103	Presence of lacquer on face of primer		0.40%	3.2	Visual
104	Distance, max. from face of primer to shoulder		0.40%	3.2	Gage
105	Body not fully assembled to head		0.40%	3.2	Visual
106	Body and head improperly staked (2 places)		0.40%	3.2	Gage
107	Pellet missing		100%	3.2	Visual
108	Presence of adhesive for 360 degrees in body groove prior to assembling to tube assembly		0.40%	3.2	Visual
109	Tube assembly improperly seated on body		0.40%	3.2	Visual
110	Sensitivity functioning		(b)	3.4.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES: (a) See sampling plans contained in paragraph 4.4.3.4.1. (b) See sampling plans contained in paragraph 4.4.3.4.2.

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Cartridge, Ignition, Prior to Assembly of Cap				9294752
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Propellant weight (before loading)		100%	3.7	4.4.3.2/Scale
102	Propellant missing		100%	3.2	Visual
Minor					
201	Foreign matter in charge		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.7	Cartridge, Ignition, After Assembly of Cap				9294752
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	RTV sealant missing, inadequate or excessive prior to or after assembly of cap		0.40%	3.2	Visual
102	Cap cocked or not completely seated		0.40%	3.2	Visual
103	Distance from shoulder to front of cap, max.		0.40%	3.2	Gage
104	Outside diameter, max.		0.40%	3.2	Gage
105	True position of outside diameter with pitch diameter of head thread		0.40%	3.2	Gage
106	Weight of propellant (after loading)	315		3.8	4.4.3.3
107	Leak test		100%	3.5	4.5.4
108	Velocity, functioning and security (proving ground)		(a)	3.6	4.5.6
109	Firing Plug missing or inverted		100%	3.2	Visual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:
(a) See paragraph 4.4.3.5 for sampling plans.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-C-48397A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9294952
4.4.2.8	Tray Assembly for Cartridge, Ignition M752A1				NEXT HIGHER ASSEMBLY 9294950
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Cartridge missing		0.40%	3.2	Visual
102	Tray broken, split or incomplete		0.40%	3.2	Visual
103	Improperly packed		0.40%	3.2	Visual
Minor					
201	Evidence of poor workmanship		0.65%	3.10	Visual

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.9	Packed Box, Fiberboard				9294950
					NEXT HIGHER ASSEMBLY 9294951
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box broken or split		0.40%	3.2	Visual
102	Incomplete pack		0.40%	3.2	Visual
103	Improperly packed		0.40%	3.2	Visual
104	Improperly sealed		0.40%	3.2	Visual
Minor					
201	Marking incomplete or incorrect		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9294951
4.4.2.10	Box, Wirebound, Before Packing				NEXT HIGHER ASSEMBLY 9294949
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside dimensions incorrect		0.40%	3.2	Gage
102	Broken, split or missing components		0.40%	3.2	Gage
Minor					
201	Marking incomplete, incorrect or illegible		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.11	Packed Wirebound Box, Before Closing				9294949
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Improperly packed		0.40%	3.2	Visual
Minor 201	Evidence of poor workmanship		0.65%	3.10	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.12	Packed Wirebound Box, After Closing	1 OF 1	9294949
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Wire closing improper		0.40%	3.2	Visual
Minor 201	Contents loose		0.65%	3.2	Manual
202	Marking incomplete, incorrect or illegible		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.10	Visual

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4.4.3 Testing.

4.4.3.1 Primer push out test. The test shall be performed as specified in 4.5.1. If one or more samples fail to comply with the requirement in accordance with the drawing, the eight hours of production represented by the samples shall be rejected. The test samples shall be discarded after the test.

4.4.3.2 Weight of propellant before loading (see dwg. 9294752). The propellant charge shall be weighed or volumetrically dispensed into a tared container, and then check-weighed 100 percent. The initial weighing and check-weighing shall be performed by different operators on different scales. Weight limits for check-weighing should be established within the applicable weight requirement, depending on the accuracy and precision of the weighing system, to assure compliance with charge weight requirements. Any charge failing to meet the applicable requirement shall be considered defective and removed from the lot.

4.4.3.2.1 Automated method. The contractor may elect to perform this operation with automated equipment and mandatory process controls. In such cases, process control plans in accordance with Data Item Description DI-P-1604 must be submitted to the Technical Agency (see 6.10) for approval of the automated operation.

4.4.3.3 Weight of propellant in cartridge assembly. (see dwg. 9294752). 315 sample cartridge assemblies shall be randomly selected from each lot for this test. If four or more samples exceed the assessed maximum weight by 5 or less grains, or if any sample exceeds the assessed maximum weight by more than 5 grains, the lot shall be rejected. If four or more samples are under the assessed minimum weight by 2 grains or less, or if any sample is under the assessed minimum weight by more than 2 grains, the lot shall be rejected. In addition, if the total number of defective samples is five or more the lot shall be rejected. Testing shall be as specified in 4.5.5.

4.4.3.3.1 Alternate procedure. Loaded ignition cartridges shall be weighed 100 percent and the net weight of propellant determined by subtracting parts tare weight from the gross weight. The net weight of propellant should compare with the original weight of propellant within $\pm$ 0.5 grain.

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4.4.3.4 Sensitivity

4.4.3.4.1 Non-functioning (see 3.4.1). Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified in 3.4.1, the sampling plan for this test shall be in accordance with MIL-STD-105D, Table II-C, Code Letter N and an AQL of 0.025 percent. The test shall be performed as specified in 4.5.3.1.

4.4.3.4.1.1 After three consecutive lots have met the criteria of 4.4.3.4.1, the sampling plan for this test shall be in accordance with MIL-STD-105D, Table II-C, Code Letter M and AQL of 0.040 percent. The test shall be performed as specified in 4.5.3.1.

4.4.3.4.2 Functioning (see 3.4.2). Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified in 3.4.2, the sampling plan for this test shall be in accordance with MIL-STD-105D, Table III-B, Code Letter P, and AQL of 0.25 percent. Any primed body, head and firing plug assembly in the randomly selected sample missing a component or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.3.2.

4.4.3.4.2.1 After three consecutive lots have met the criteria of 4.4.3.4.2, the sampling plan for this test shall be in accordance with MIL-STD-105D, Table III-B, Code Letter N and AQL of 0.25 percent. Any primed body, head and firing plug assembly in the randomly selected sample missing a com-ponent or with an incorrectly assembled component which will result in an end item misfire shall be considered defective and shall be included in the rejection criteria. The test shall be performed as specified in 4.5.3.2.

4.4.3.4.3 Safety drop test. Twenty five test assemblies consisting of the primed body head, firing plug, flash tube/pellet assembly, tube and caps (assembled without M9 in accordance with dwg. 9294752) shall be randomly selected for this test. If one or more of the test assemblies function when tested as specified in 4.5.3.3, the primer lot shall be rejected.

4.4.3.5 Velocity, functioning and security. (see 3.6)

4.4.3.5.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the

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applicable requirements specified, the sampling plan and procedure for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter N and an AQL of 1.0 percent. In addition, if the standard deviation of the sample exceeds the lot requirement, the lot shall be rejected. If one or more misfires occurs (see para. 3.7.1) or if the metal parts security requirement is not met (see para. 3.7.3), the ignition cartridge lot shall be rejected. The test shall be performed as specified in para 4.5.6.

4.4.3.5.2 After three consecutive lots have met the criteria of 4.4.3.5.1, the sampling plan for this test shall be in accordance with MIL-STD-414, Section B, Table B-3, Code Letter K, and an AQL of 1.0 percent. In addition, if the standard deviation of the sample exceeds the applicable lot requirement, the lot shall be rejected. If one or more misfire occurs (see para. 3.7.1) or if the metal parts security requirement is not met (see para 3.7.3), the ignition cartridge lot shall be rejected. The test shall be performed as specified in paragraph 4.5.6.

4.4.4 Inspection Equipment. The inspection equipment required to perform the examinations and tests prescribed in this specification is identified either directly or by reference, in the "Paragraph Reference/Inspection Method" column of the Quality Conformance Inspection tables herein. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

4.5 Methods of inspection

PRECAUTION - This specification covers sampling and testing of chemical, toxic or explosive materials which are potentially hazardous to personnel. Accordingly, it is emphasized that all applicable safety rules, regulations and procedures must be followed in handling and processing these materials.

4.5.1. Primer push out test. The swaged primer shall remain securely seated while subjected to the minimum push out force as specified on drawing 12630551. The equipment and procedures used in this push out test shall be submitted to the Government for approval. The approved equipment shall be periodically calibrated.

4.5.2 Moisture content of black powder. The moisture content of black powder shall be determined in accordance with the procedure specified in Specification MIL-P-223. A sample of 25 black powder pellets shall be randomly selected from each eight

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hours production. Any failure to comply with the requirement cited in paragraph 3.3 shall be cause for rejection of the eight hours production represented by the sample. If loading has begun, ignition cartridges loaded with the non-conforming black powder shall be rejected. The contractor shall submit a quality inspection defect report in accordance with Data Item Description (DI-R-1721) giving the detailed test results.

4.5.3 Sensitivity

4.5.3.1 Non-functioning. The primed body, head and firing plug, as assembled, shall be assembled to a suitable fixture. The firing pin of the test fixture shall be adjusted so that impact will be on the center of the firing plug of the primer. A weight of 2.95 pounds, mimimum, shall be dropped onto the firing pin from a height of 3/16 inch, minimum, as if to function the primer. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which functions shall be classed defective. Metal parts used in this test shall not be returned to the lot.

4.5.3.2 Functioning. The primed body head and firing plug, as assembled, shall be assembled to an appropriate test fixture. The firing pin of the test fixture shall be adjusted so that impact will be 0.05 inch off center of the firing plug of the primer. A weight of 2.95 pound, maximum, shall be dropped on the firing pin to function the primer from a height of 1 3/4 inches, maximum. The height of drop shall be measured as the height from the top of the firing pin to the bottom of the weight. Any primer which fails to function shall be classed defective. Metal parts used in this test shall not be returned to the lot.

4.5.3.3 Safety drop test. The test assemblies shall be assembled into inert-loaded, standard 81mm cartridges weighted to 10.35 pounds. Each cartridge shall be dropped twice from a height of five feet on an anvil surface. For the first drop, the cartridge shall be in the base (fin) down orientation. For the second drop, the cartridge shall be in the nose (fuze) down orientation. The drop height shall be measured from the lowest point of the cartridge orientation to the ground. Any test assembly which functions as a result of the drop shall be classed defective. All questionable test assemblies shall be down loaded to verify primer/black powder pellet functioning or non-functioning. All components used for this test shall not be returned to their respective lots.

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4.5.4 Leak test. The ignition cartridges shall be tested 100% in accordance with drawing 9278622. Any ignition cartridge which fails to comply with the applicable requirement shall be classed defective and removed from the lot. The contractor shall submit a quality inspection defect report in accordance with Data Item Description DI-R-1721 giving the detailed test results.

4.5.5 Weight of propellant. The sample cartridge assemblies shall be cut open using a method that will not generate chips and all propellant removed and placed on the pan of a precision analytical balance. The weight of propellant from each sample shall be determined to the nearest 0.01 grain.

4.5.6 Velocity, functioning and security (proving ground test. The test shall be performed at a Government proving ground in accordance with the applicable USATECOM Acceptance Test Procedure.

The sample ignition cartridges shall be assembled to 81MM cartridges, (Weighed to 8.9 ± 0.02 lbs. for M853) for which they were manufactured. The complete round, less propelling charges, shall be fired from a M252 81mm mortar at an elevation of forty-five degrees.

Reference rounds shall be fired without propelling charges for the purpose of velocity correcton. At least one reference round should be fired for every three test rounds.

Individual velocities shall be recorded on the firing record for both reference and test rounds. Velocity standard deviations and arithmetic averages shall be calculated and recorded. Temperature of the day and time of firing shall also be recorded. In the event of a misfire, the misfired ignition cartridges shall not be disassembled by the proving ground but shall be forwarded to the responsible technical activity.

4.5.6.1 Refiring. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, additional cartridges as required shall be tested.

4.5.7 Propellant charge weight assessment. (To be conducted at a U.S. Government Proving Ground.) Ten test ignition cartridges shall be loaded at charge weight established during the propellant assessment in accordance with MIL-P-60398 with a tolerance of plus or minus 0.5 grains. The ignition cartridges shall be assembled to 81mm production rounds. Rounds shall be inert loaded to achieve a

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total weight of 8.9 pounds plus or minus .02 pounds for M853. The rounds shall be conditioned at 70°F (+2.5°F) for 24 hours prior to firing and shall be maintained at this temperature until fired. The rounds shall be fired from a M252 81mm mortar at an elevation of forty-five degrees. Calibration rounds shall be fired alternately for velocity corrections. The corrected velocities of the group shall be compared to the established charge velocity. If the corrected velocity of the test rounds exceeds two percent of the expected muzzle velocity, use of the propellant lot shall be referred. Test results shall be forwarded to the technical agency for evaluation (Commander, US Army Armament, Develoment and Engineering Center, ATTN: SMCAR-AEE-BP, AMSMC-QAR-R, AMSMC-QAT-B, Dover, NJ 07801).

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be as specified on Drawing 9294952 and 9294950.

5.2 Packing and marking. Packing and marking shall be as specified on Drawings 9294951 and 9294949.

5.3 Transportation regulations. 49 CFR 71-90 Interstate Commerce Commission Rules and Regulations for the Transportation of Explosive and Other Dangerous Articles shall be complied with.

6. NOTES

6.1 Intended use. The M752A1 ignition cartridge covered by this specification is intended for use on the Cartridges, 81mm, Illuminating M853 and Smoke M819.

6.2 Ordering data. See MIL-A-48078 (AR).

6.3 Submission of inspection equipment for design approvals. See MIL-A-48078. Submit designs as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-I, Dover, NJ 07801-5001.

6.4 New material test sample requirements. All requests for permission to use materials other than those specified on applicable drawings or specifications should be accompanied by a sample of the proposed substitute material to permit evaluation by AMCCOM. Prior to approval, all substitute materials must be tested for compatibility with M9 Propellant in addition to such testing required to evaluate the performance of the proposed substitute.

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6.5 Submission of test data. In addition to the normal distribution of records, when the ignition cartridge is procured by the Department of the Army, one copy of the formal firing report containing velocity test data and lot acceptability shall be sent to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-B, Dover, NJ 07801-5001.

6.6 Check test for deterioration of ignition cartridge, M752A1. If the total elapsed time between original acceptance of the ignition cartridge lot and the assembly of that lot into the complete round exceeds two years, or if the ignition cartridges have been subjected to adverse conditions for any period of time since previous tests, the ignition cartridges shall be subjected to, and must satisfactorily comply with the following tests immediately before the ignition cartridges are assembled into the complete round.

6.6.1 Non-functioning. The requirement shall be as specified in para. 3.4.1. The sampling plan for this test shall be as specified in para. 4.4.3.4.1. The test shall be performed as specified in 4.5.3.1.

6.6.2 Functioning. The requirement shall be as specified in para. 3.4.2. The sampling plan for this test shall be as specified in para. 4.4.3.4.2. The test shall be performed as specified in 4.5.3.2.

6.6.3 Moisture content of black powder pellet. Twenty-five sample black powder pellets shall be selected from the ignition cartridge lot. The requirement shall be as specified in para. 3.3. The test shall be performed as specified in para. 4.5.2.

6.7 Check test for deterioration of primer. If the total elapsed time between original acceptance of the primer lot and the assembly of that lot into the ignition cartridge exceeds two years, or if the primers have been subjected to adverse conditions for any period of time since previous test, the primer lot shall be subjected to and must satisfactorily pass the check test for deterioration specified in Specification MIL-P-46425 immediately before the primer lot is assembled into the ignition cartridges.

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6.8 Data requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data Requirement	Applicable DID
4.5.2, 4.5.4	Quality Inspection Report	DI-R-1721
4.4.3.2.1	Manufacturing Methods Reports (Modified)	DI-P-1604

(Copies of data item descriptions required by the contractor in connection with the specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.9 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal, Picatinny Arsenal or ARRADCOM drawings. Technical data originally prepared by these activities is now under the cognizance of ARDEC.

6.10 Submission of process control plans for approval. Submit plans as required to Commander, US Army Armament Research, Development and Engineering Center, ATTN: AMSMC-QAT-B, Dover, NJ 07801-5001.

6.11 Subject term (key word) listing.

Cartridge, ignition
Functioning
Leak test
Non-functioning
Pellet
Security
Sensitivity
Velocity

6.12 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Preparing activity:
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(Project 1315-A394)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-48397A	CARTRIDGE, IGNITION, M752A1 LOADING, ASSEMBLING AND PACKING

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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